

BIOLOGY 101 · WEEK 2 QUIZ

Biology 101 — Week 2

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. How many rooms does the heart have, and what are the two kinds?

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- A. Four: two atria that receive blood and two ventricles that pump it out.
- B. Four: two for oxygen, one for carbon dioxide, and one for plasma.
- C. Three: one room for each pipe of the umbilical cord.
- D. Two: one atrium for red blood and one ventricle for blue blood.

2. Where does the RIGHT side of the heart send the blood it pumps?

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- A. To the left side of the heart, through the ductus.
- B. To the whole body, to deliver oxygen to every cell.
- C. To the lungs, to drop off CO₂ and pick up oxygen.
- D. To the liver, where new blood is made.

3. A newborn's heart beats about 140 times a minute. About how many beats is that in a day?

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- A. About 8,400
- B. About 20,000
- C. About 2,000,000
- D. About 200,000

4. Which part of the blood carries oxygen?

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- A. Red blood cells, using the hemoglobin packed inside them.
- B. White blood cells, which carry oxygen to fight germs.
- C. Plasma, the watery part that dissolves the oxygen.
- D. Platelets, which carry oxygen to wounds to heal them.

5. About how much blood does a 1-kilogram preemie have in its whole body?

- ☐ A. About 8 mL – roughly two teaspoons.
- ☐ B. About 80 mL – roughly a third of a cup.
- ☐ C. About 5 liters – the same as an adult.
- ☐ D. About 800 mL – roughly a big soda bottle.

6. What is the ductus arteriosus?

- ☐ A. The vein in the umbilical cord that brings oxygen-rich blood from the placenta.
- ☐ B. A tube doctors place in a preemie's artery to measure blood pressure.
- ☐ C. The one-way door between the right atrium and the right ventricle.
- ☐ D. A fetal bypass pipe that lets blood skip the unused lungs; it should close after birth.

7. Why is an open ductus (a PDA) a problem for a preemie?

- ☐ A. Blood leaks back into the lungs, flooding them, while the body gets less than its share.
- ☐ B. Blood skips the lungs entirely, so the baby's oxygen level drops all the way down to zero.
- ☐ C. The umbilical cord stays attached to the placenta, so the baby cannot be moved or held.
- ☐ D. The heart's valves stop closing properly, so the 'lub-dub' sound disappears entirely.

8. What did William Harvey show in 1628 – and how?

- ☐ A. That blood is made in the liver, using a microscope and a drop of blood.
- ☐ B. That babies' hearts have a shortcut, using an ultrasound of a newborn.
- ☐ C. That blood goes around in a loop, using arithmetic and the one-way valves in veins.
- ☐ D. That the heart runs on electricity, using stickers on the chest of a patient.

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)